COMPUTO

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How to Train Your Intercept

A Comparison of Intercept Update Strategies in Coordinate Descent

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Abstract

Training intercepts in regularized regression is a common problem, yet it is often overlooked in the literature. This document explores different strategies for updating intercepts in coordinate descent algorithms, comparing gradient, Newton, and exact strategies.

Keywords: intercepts, coordinate descent, optimization, regularization

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1 Introduction

Intercept updating is a overlooked issue in the field of statistical learning. For regularized regression, coordinate descent is the dominating algorithm, yet it is not clear how to best update the intercept term. Implementations differ in how they handle this, for instance skglm uses a gradient step, LIBLINEAR uses a Newton step, and blitz uses an exact update (based on Newton steps).

References

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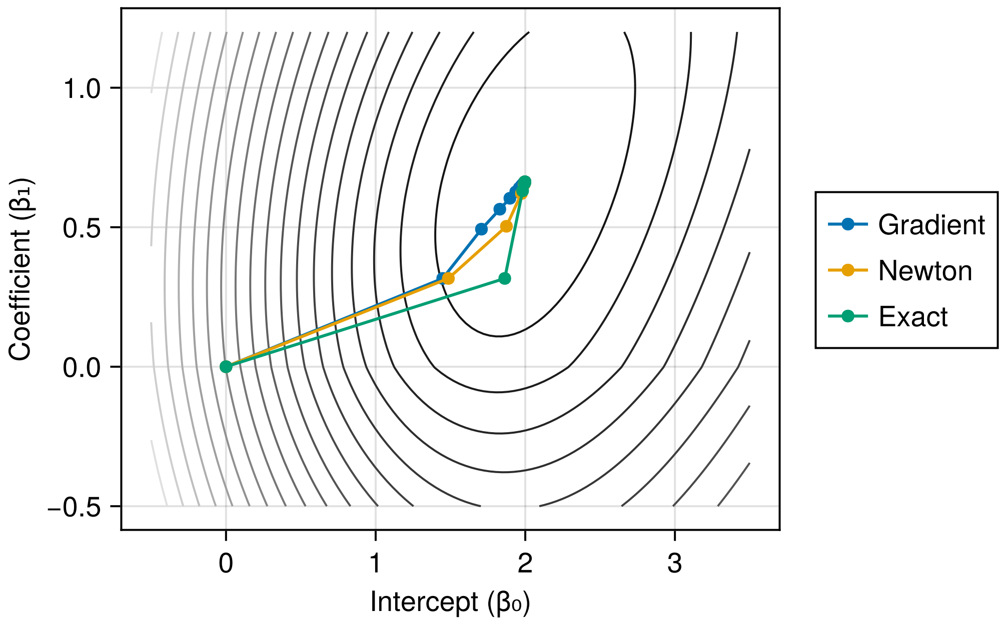

Figure 1: Parametric Plos